Supplementary Material
How Hard Is the Attack? A Complexity-Theoretic Systematization of Machine Learning Security

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1. Classifying the corpus

This section reports the corpus in two layers. Tables 2 to 4 cross-tabulate features against one another, with works in the cells by citation number. Tables 5 and 6 give one row per work, so that every cell above is traceable to the paper behind it. Every value comes from a fixed vocabulary settled before the pass began.

The cross-tabulations are the layer that carries an argument, because in them an empty cell is a combination nobody has published. The value `none` is an answer and not a missing entry, recording that a work constrains no party at all. Of the 181 works classified so far, 32 state no resource bound on any party, and 0 fall outside the frame in the sense that deleting every resource restriction leaves the main claim unchanged.

This is a partial pass. 181 of the corpus's works are classified. The remainder are not yet placed.

TABLE 1: The corpus by pipeline phase (rows) against access level (columns). Each cell lists every work with that combination, by citation number. An empty cell is a combination no work in the corpus occupies.

phase	**hard label**	**scores**	**black box**	**white box**	**weights**	**gradients**	**channel**	**none**	***unassigned***	Σ
train time	[62]		[105], [146]	[68], [71], [137], [140], [166], [185], [195], [197]		[28]	[192]	[4], [12], [17], [22], [34], [36], [46], [70], [72], [80], [131], [160], [162], [168], [190], [191]		29
test time	[92]		[32], [57], [96], [97], [101], [158], [167], [174]	[6], [13], [49], [50], [55], [65], [66], [98], [143], [144], [173], [176], [177], [181], [184], [187]			[114]	[10], [40], [43], [51], [78], [85], [93], [108], [136], [142], [150], [151], [156], [170], [199]		41
post hoc		[16], [53], [54], [61], [86], [117], [122], [159], [172], [180], [196]	[52], [74], [77], [109], [124], [152], [171], [188]	[25], [58], [59], [95], [100], [126], [130], [134], [145], [157], [163], [182], [186], [193]	[121]	[127]		[63], [90], [149], [153], [179]	[164]	41
n/a		[79]						[1], [2], [3], [5], [8], [9], [11], [14], [15], [20], [21], [23], [24], [26], [27], [33], [35], [37], [38], [39], [41], [42], [44], [45], [48], [56], [60], [64], [67], [69], [73], [75], [76], [81], [82], [83], [84], [87], [88], [89], [91], [94], [99], [102], [103], [104], [106], [107], [110], [116], [118], [119], [120], [123], [129], [133], [135], [138], [141], [147], [148], [154], [161], [169], [175], [178], [189], [194], [198]		70
Σ	2	12	18	38	1	2	2	105	1	181

TABLE 2: The corpus by constrained party (rows) against resource unit (columns). Each cell lists every work with that combination, by citation number. An empty cell is a combination no work in the corpus occupies. A dagger marks works where the resource appears only as a side condition rather than as a parameter the work assesses on purpose.

party	time	circuit size	queries	samples	memory	algo class	parameters	perturbation	corruption	rounds	privacy budget	circuit depth	success p
attacker	[11], [26], [33], [45], [62], [63], [71], [74], [79], [81], [92], [100], [106], [117], [118], [126], [138], [147], [148], [180], [185], [192], [28], [54], [95], [174], [195], [197]†	[1], [23], [38], [48], [99], [101]	[5], [42], [52], [57], [122], [145], [53], [61], [96], [152], [172]†	[121], [164]†	[24], [83], [84], [110]	[21], [60], [89], [124], [189]		[6], [13], [50], [65], [66], [78], [93], [108], [114], [136], [142], [150], [151], [156], [170], [173], [176], [177], [181], [184], [187], [49], [55]†	[137], [140], [146], [160], [168]			[107]	
learner	[15], [20], [34], [36], [37], [40], [44], [69], [72], [76], [80], [119], [123], [135], [141], [178], [198]	[103]	[22]†	[10], [51], [70], [73], [75], [82], [88], [143], [144], [190]	[162]	[2], [3], [8], [17], [43], [56], [67], [91], [94], [104], [131], [175], [191]	[46], [102]						
auditor	[58], [68], [105], [109], [130], [32], [196]†			[153], [199]		[35], [77], [116]							
mechanism none	[27], [161]				[87], [133]	[194]†					[4], [163]†		
Σ	54	7	12	14	7	22	2	23	5	0	2	1	0

TABLE 3: The corpus by resource unit (rows) against tightness of the bound (columns). Each cell lists every work with that combination, by citation number. An empty cell is a combination no work in the corpus occupies.

unit	**unstated**	**poly**	**quasi-poly**	**class**	**budget**	**conditional**	**exponent**	**measured**	Σ
time	[126], [196]	[27], [37], [40], [58], [62], [63], [68], [72], [79], [80], [81], [92], [100], [105], [106], [109], [117], [118], [119], [123], [130], [141], [147], [148], [178], [180], [185], [192], [198]	[34]		[161], [197]	[11], [15], [20], [36], [44], [69]	[26], [32], [33], [45], [76], [135], [138], [195]	[28], [54], [71], [74], [95], [174]	54
circuit size		[101]	[1], [48], [99]		[38]	[103]	[23]		7
queries					[22], [52], [96], [145], [152]		[5], [42], [57], [122]	[53], [61], [172]	12
samples				[70], [190]	[10], [51], [143], [144], [199]		[73], [75], [82], [88], [121], [153]	[164]	14
memory					[162]	[87]	[24], [83], [84], [110], [133]		7
algo class				[2], [3], [8], [17], [21], [35], [43], [56], [60], [67], [77], [89], [91], [94], [104], [116], [124], [131], [175], [189], [191], [194]					22
parameters					[46], [102]				2
perturbation					[6], [13], [49], [50], [55], [65], [66], [78], [93], [108], [114], [136], [142], [150], [151], [156], [170], [173], [176], [177], [181], [184], [187]				23
corruption					[137], [140], [146], [168]			[160]	5
rounds									0
privacy budget					[4], [163]				2
circuit depth							[107]		1
success prob									0
none	[9], [12], [14], [16], [25], [39], [41], [59], [64], [85], [86], [90], [97], [98], [120], [127], [129], [134], [149], [154], [157], [158], [159], [166], [167], [169], [171], [179], [182], [186], [188], [193]								32
Σ	34	30	4	24	45	8	25	11	181

TABLE 4: The corpus by attack family (rows) against constrained party (columns). Each cell lists every work with that combination, by citation number. An empty cell is a combination no work in the corpus occupies.

family	**attacker**	**learner**	**auditor**	**mechanism**	**none**	Σ
membership	[54], [100], [121], [172], [185]		[196]		[16], [86], [90], [149], [159], [193]	12
attribute						0
reconstruct	[28], [63], [79], [117], [122], [145]				[25], [59], [134], [188]	10
distribution	[180], [192]					2
extract	[52], [53], [61], [74], [124]				[171]	6
invert						0
forge	[71], [84], [95], [126], [197]				[127], [166], [182]	8
evade	[6], [13], [49], [50], [55], [65], [66], [78], [92], [93], [96], [101], [108], [114], [136], [142], [150], [151], [156], [170], [173], [174], [176], [177], [181], [184], [187]	[10], [40], [43], [51], [80], [143], [144]	[32]		[97], [98]	37
evade audit	[57], [195]		[153]		[158], [179]	5
inject			[199]			1
poison	[137], [140], [146], [168]	[22], [70], [72], [190]				8
backdoor	[152], [160]		[58], [68], [105], [109], [130]		[12], [157], [186]	10
multiple					[129], [154]	2
none	[1], [5], [11], [21], [23], [24], [26], [33], [38], [42], [45], [48], [60], [62], [81], [83], [89], [99], [106], [107], [110], [118], [138], [147], [148], [189]	[2], [3], [8], [15], [17], [20], [34], [36], [37], [44], [46], [56], [67], [69], [73], [75], [76], [82], [88], [91], [94], [102], [103], [104], [119], [123], [131], [135], [141], [162], [175], [178], [191], [198]	[35], [77], [116]	[4], [27], [87], [133], [161], [194]	[9], [14], [39], [41], [64], [120], [167], [169]	77
unassigned	[164]			[163]	[85]	3
Σ	85	45	12	7	32	181

TABLE 5: Per-work reference layer: attack scenario and threat model.

#	Target	Family	Phase	Access	Write	Frame
[90]	train data	membership	post hoc	none	read only	none
[159]	train data	membership	post hoc	scores	read only	none
[86]	train data	membership	post hoc	scores	read only	none
[185]	train data	membership	train time	white box	corpus	compute
[121]	train data	membership	post hoc	weights	read only	statistical
[172]	train data	membership	post hoc	scores	read only	compute
[196]	train data	membership	post hoc	scores	read only	compute
[54]	train data	membership	post hoc	scores	read only	compute
[16]	train data	membership	post hoc	scores	read only	none
[193]	train data	membership	post hoc	white box	read only	none
[149]	train data	membership	post hoc	none	none	none
[100]	train data	membership	post hoc	white box	read only	compute
[59]	train data	reconstruct	post hoc	white box	read only	none
[63]	train data	reconstruct	post hoc	none	read only	compute
[79]	train data	reconstruct	n/a	scores	read only	compute
[134]	train data	reconstruct	post hoc	white box	read only	none
[28]	train data	reconstruct	train time	gradients	read only	compute
[122]	train data	reconstruct	post hoc	scores	read only	compute
[145]	train data	reconstruct	post hoc	white box	read only	compute
[25]	train data	reconstruct	post hoc	white box	read only	none
[188]	train data	reconstruct	post hoc	black box	read only	none
[117]	train data	reconstruct	post hoc	scores	read only	compute
[180]	train data	distribution	post hoc	scores	read only	compute
[192]	train data	distribution	train time	channel	corpus	compute
[52]	parameters	extract	post hoc	black box	read only	compute
[171]	parameters	extract	post hoc	black box	read only	none
[74]	parameters	extract	post hoc	black box	read only	compute
[61]	parameters	extract	post hoc	scores	read only	compute
[124]	parameters	extract	post hoc	black box	read only	compute
[53]	parameters	extract	post hoc	scores	read only	compute
[182]	training run	forge	post hoc	white box	corpus	none
[84]	none	forge	n/a	none	none	compute
[127]	train data	forge	post hoc	gradients	corpus	none
[197]	training run	forge	train time	white box	model	compute
[95]	training run	forge	post hoc	white box	corpus	compute
[71]	training run	forge	train time	white box	model	compute
[126]	training run	forge	post hoc	white box	model	compute
[166]	training run	forge	train time	white box	corpus	none
[181]	outputs	evade	test time	white box	read only	capability
[55]	outputs	evade	test time	white box	read only	capability
[170]	outputs	evade	test time	none	read only	capability
[43]	outputs	evade	test time	none	read only	compute
[40]	outputs	evade	test time	none	none	compute
[108]	outputs	evade	test time	none	read only	capability
[93]	outputs	evade	test time	none	read only	capability
[10]	outputs	evade	test time	none	read only	statistical
[136]	outputs	evade	test time	none	read only	capability
[65]	outputs	evade	test time	white box	read only	capability
[151]	outputs	evade	test time	none	read only	capability
[92]	outputs	evade	test time	hard label	read only	compute
[101]	outputs	evade	test time	black box	read only	compute
[142]	outputs	evade	test time	none	read only	capability
[156]	outputs	evade	test time	none	read only	capability
[150]	outputs	evade	test time	none	read only	capability
[32]	outputs	evade	test time	black box	read only	compute
[98]	outputs	evade	test time	white box	read only	none
[97]	outputs	evade	test time	black box	read only	none
[78]	outputs	evade	test time	none	read only	capability

Table 5, continued

#	Target	Family	Phase	Access	Write	Frame
[66]	outputs	evade	test time	white box	read only	capability
[6]	outputs	evade	test time	white box	read only	capability
[49]	outputs	evade	test time	white box	read only	capability
[51]	outputs	evade	test time	none	read only	statistical
[187]	outputs	evade	test time	white box	read only	capability
[177]	outputs	evade	test time	white box	read only	capability
[173]	outputs	evade	test time	white box	read only	capability
[144]	outputs	evade	test time	white box	read only	statistical
[50]	outputs	evade	test time	white box	read only	capability
[184]	outputs	evade	test time	white box	read only	capability
[114]	outputs	evade	test time	channel	channel	capability
[176]	outputs	evade	test time	white box	read only	capability
[96]	outputs	evade	test time	black box	read only	compute
[174]	outputs	evade	test time	black box	read only	compute
[13]	outputs	evade	test time	white box	read only	capability
[143]	outputs	evade	test time	white box	none	statistical
[80]	none	evade	train time	none	none	compute
[179]	training run	evade audit	post hoc	none	none	none
[153]	training run	evade audit	post hoc	none	responses	statistical
[158]	outputs	evade audit	test time	black box	read only	none
[57]	outputs	evade audit	test time	black box	responses	compute
[195]	training run	evade audit	train time	white box	model	compute
[199]	outputs	inject	test time	none	channel	statistical
[168]	train data	poison	train time	none	corpus	capability
[140]	train data	poison	train time	white box	corpus	capability
[137]	train data	poison	train time	white box	corpus	capability
[146]	train data	poison	train time	black box	corpus	capability
[72]	train data	poison	train time	none	corpus	compute
[70]	train data	poison	train time	none	corpus	compute
[190]	train data	poison	train time	none	corpus	compute
[22]	train data	poison	train time	none	corpus	compute
[58]	parameters	backdoor	post hoc	white box	read only	compute
[157]	parameters	backdoor	post hoc	white box	read only	none
[152]	outputs	backdoor	post hoc	black box	read only	compute
[186]	parameters	backdoor	post hoc	white box	read only	none
[130]	parameters	backdoor	post hoc	white box	model	compute
[109]	parameters	backdoor	post hoc	black box	read only	compute
[105]	parameters	backdoor	train time	black box	model	compute
[12]	train data	backdoor	train time	none	corpus	none
[68]	parameters	backdoor	train time	white box	model	compute
[160]	parameters	backdoor	train time	none	corpus	capability
[129]	none	multiple	n/a	none	none	none
[154]	none	multiple	n/a	none	none	none
[133]	none	none	n/a	none	none	compute
[9]	none	none	n/a	none	none	none
[45]	none	none	n/a	none	none	compute
[76]	none	none	n/a	none	none	compute
[17]	none	none	train time	none	none	compute
[35]	none	none	n/a	none	none	compute
[38]	none	none	n/a	none	none	compute
[110]	none	none	n/a	none	none	compute
[21]	none	none	n/a	none	none	compute
[44]	none	none	n/a	none	none	compute
[24]	none	none	n/a	none	none	compute
[60]	train data	none	n/a	none	none	compute
[118]	none	none	n/a	none	none	compute
[148]	none	none	n/a	none	none	compute
[147]	train data	none	n/a	none	none	compute
[62]	training run	none	train time	hard label	read only	compute

Table 5, continued

#	Target	Family	Phase	Access	Write	Frame
[1]	none	none	n/a	none	none	compute
[4]	train data	none	train time	none	none	capability
[37]	none	none	n/a	none	none	compute
[69]	none	none	n/a	none	none	compute
[14]	none	none	n/a	none	none	none
[87]	none	none	n/a	none	none	compute
[161]	train data	none	n/a	none	none	compute
[94]	none	none	n/a	none	none	compute
[89]	none	none	n/a	none	none	compute
[36]	none	none	train time	none	none	compute
[15]	none	none	n/a	none	none	compute
[5]	none	none	n/a	none	none	compute
[26]	none	none	n/a	none	none	compute
[23]	none	none	n/a	none	none	compute
[11]	none	none	n/a	none	none	compute
[167]	outputs	none	test time	black box	corpus	none
[103]	none	none	n/a	none	none	compute
[104]	none	none	n/a	none	none	compute
[82]	none	none	n/a	none	none	compute
[75]	none	none	n/a	none	none	compute
[141]	none	none	n/a	none	none	compute
[178]	none	none	n/a	none	none	compute
[120]	none	none	n/a	none	none	none
[123]	none	none	n/a	none	none	compute
[194]	none	none	n/a	none	none	compute
[8]	none	none	n/a	none	none	compute
[77]	outputs	none	post hoc	black box	read only	compute
[39]	none	none	n/a	none	none	none
[91]	none	none	n/a	none	none	compute
[169]	none	none	n/a	none	none	none
[3]	none	none	n/a	none	none	compute
[67]	none	none	n/a	none	none	compute
[119]	none	none	n/a	none	none	compute
[106]	none	none	n/a	none	none	compute
[56]	none	none	n/a	none	none	compute
[175]	none	none	n/a	none	none	compute
[191]	none	none	train time	none	none	compute
[64]	none	none	n/a	none	none	none
[2]	none	none	n/a	none	none	compute
[42]	none	none	n/a	none	none	compute
[189]	none	none	n/a	none	none	compute
[73]	none	none	n/a	none	none	compute
[102]	none	none	n/a	none	none	compute
[33]	none	none	n/a	none	none	compute
[138]	none	none	n/a	none	none	compute
[48]	none	none	n/a	none	none	compute
[99]	none	none	n/a	none	none	compute
[135]	none	none	n/a	none	none	compute
[34]	none	none	train time	none	none	compute
[162]	none	none	train time	none	none	compute
[20]	none	none	n/a	none	none	compute
[88]	none	none	n/a	none	none	compute
[116]	none	none	n/a	none	none	compute
[27]	none	none	n/a	none	none	compute
[107]	none	none	n/a	none	none	compute
[81]	none	none	n/a	none	none	compute
[83]	none	none	n/a	none	none	compute
[46]	none	none	train time	none	none	compute
[131]	train data	none	train time	none	none	compute

Table 5, continued

#	Target	Family	Phase	Access	Write	Frame
[198]	none	none	n/a	none	none	compute
[41]	none	none	n/a	none	none	none
[163]	train data	·	post hoc	white box	read only	capability
[85]	outputs	·	test time	none	responses	none
[164]	train data	·	post hoc	·	read only	statistical

TABLE 6: Per-work reference layer: adversary assumptions and claim shape. Same rows and same order as Table 5.

#	Party	Unit	Tightness	Baseline	2nd party	2nd notion	Assessed	Quantifier	Antecedent
[90]	none	none	unstated	stated	none	none	·	empirical	stated
[159]	none	none	unstated	stated	none	none	·	empirical	stated
[86]	none	none	unstated	stated	none	none	·	empirical	stated
[185]	attacker	time	poly	stated	none	none	explicit	conditional	stated
[121]	attacker	samples	exponent	stated	samples	·	explicit	universal	stated
[172]	attacker	queries	measured	stated	none	none	implicit	empirical	stated
[196]	auditor	time	unstated	stated	none	none	implicit	conditional	stated
[54]	attacker	time	measured	stated	none	none	implicit	empirical	stated
[16]	none	none	unstated	stated	none	none	·	empirical	stated
[193]	none	none	unstated	stated	none	none	·	conditional	stated
[149]	none	none	unstated	stated	none	none	·	empirical	stated
[100]	attacker	time	poly	stated	none	queries	explicit	conditional	stated
[59]	none	none	unstated	stated	none	·	·	universal	stated
[63]	attacker	time	poly	stated	time	none	explicit	conditional	stated
[79]	attacker	time	poly	absent	perturbation	queries	explicit	universal	stated
[134]	none	none	unstated	stated	none	·	·	existential	stated
[28]	attacker	time	measured	absent	none	none	implicit	empirical	stated
[122]	attacker	queries	exponent	stated	samples	none	explicit	conditional	stated
[145]	attacker	queries	budget	absent	privacy budget	none	explicit	universal	stated
[25]	none	none	unstated	stated	none	none	·	conditional	stated
[188]	none	none	unstated	n/a	none	none	·	empirical	stated
[117]	attacker	time	poly	stated	none	queries	explicit	conditional	stated
[180]	attacker	time	poly	stated	·	none	explicit	conditional	stated
[192]	attacker	time	poly	stated	time	none	explicit	conditional	stated
[52]	attacker	queries	budget	absent	none	none	explicit	conditional	stated
[171]	none	none	unstated	absent	none	none	·	empirical	stated
[74]	attacker	time	measured	stated	none	none	explicit	empirical	stated
[61]	attacker	queries	measured	absent	none	none	implicit	empirical	stated
[124]	attacker	algo class	class	absent	none	none	explicit	existential	stated
[53]	attacker	queries	measured	stated	none	time	implicit	empirical	stated
[182]	none	none	unstated	n/a	none	none	·	existential	stated
[84]	attacker	memory	exponent	stated	memory	none	explicit	universal	stated
[127]	none	none	unstated	absent	none	none	·	empirical	stated
[197]	attacker	time	budget	n/a	time	none	implicit	conditional	stated
[95]	attacker	time	measured	stated	none	none	implicit	empirical	stated
[71]	attacker	time	measured	stated	none	none	explicit	empirical	stated
[126]	attacker	time	unstated	absent	none	none	explicit	conditional	stated
[166]	none	none	unstated	n/a	none	none	·	empirical	stated
[181]	attacker	perturbation	budget	stated	none	none	explicit	existential	none
[55]	attacker	perturbation	budget	n/a	none	none	implicit	empirical	none
[170]	attacker	perturbation	budget	n/a	none	none	·	conditional	stated
[43]	learner	algo class	class	n/a	perturbation	·	explicit	existential	stated
[40]	learner	time	poly	stated	perturbation	none	explicit	conditional	stated
[108]	attacker	perturbation	budget	n/a	none	none	explicit	existential	stated
[93]	attacker	perturbation	budget	n/a	none	none	·	conditional	stated
[10]	learner	samples	budget	n/a	none	·	explicit	universal	stated
[136]	attacker	perturbation	budget	n/a	none	·	explicit	universal	stated
[65]	attacker	perturbation	budget	n/a	none	none	·	universal	stated
[151]	attacker	perturbation	budget	n/a	none	none	·	conditional	stated
[92]	attacker	time	poly	n/a	none	none	·	universal	stated
[101]	attacker	circuit size	poly	n/a	none	none	·	existential	stated
[142]	attacker	perturbation	budget	n/a	none	none	explicit	conditional	stated
[156]	attacker	perturbation	budget	n/a	none	none	·	conditional	stated
[150]	attacker	perturbation	budget	n/a	none	none	explicit	empirical	stated
[32]	auditor	time	exponent	stated	none	·	implicit	existential	stated
[98]	none	none	unstated	stated	none	·	·	conditional	stated
[97]	none	none	unstated	n/a	none	none	·	empirical	stated

Table 6, continued

#	Party	Unit	Tightness	Baseline	2nd party	2nd notion	Assessed	Quantifier	Antecedent
[78]	attacker	perturbation	budget	n/a	samples	none	explicit	conditional	stated
[66]	attacker	perturbation	budget	n/a	none	none	explicit	conditional	stated
[6]	attacker	perturbation	budget	stated	none	none	explicit	existential	none
[49]	attacker	perturbation	budget	n/a	none	none	implicit	empirical	none
[51]	learner	samples	budget	n/a	none	none	explicit	universal	stated
[187]	attacker	perturbation	budget	n/a	none	none	·	universal	stated
[177]	attacker	perturbation	budget	n/a	none	none	·	empirical	stated
[173]	attacker	perturbation	budget	n/a	samples	none	explicit	existential	stated
[144]	learner	samples	budget	stated	none	none	explicit	universal	stated
[50]	attacker	perturbation	budget	stated	none	time	explicit	empirical	stated
[184]	attacker	perturbation	budget	stated	none	none	explicit	existential	stated
[114]	attacker	perturbation	budget	n/a	samples	none	explicit	conditional	stated
[176]	attacker	perturbation	budget	n/a	samples	none	explicit	conditional	stated
[96]	attacker	queries	budget	stated	none	algo class	implicit	existential	none
[174]	attacker	time	measured	n/a	none	none	implicit	empirical	none
[13]	attacker	perturbation	budget	n/a	samples	none	explicit	universal	stated
[143]	learner	samples	budget	n/a	none	none	explicit	universal	stated
[80]	learner	time	poly	n/a	perturbation	none	explicit	conditional	stated
[179]	none	none	unstated	n/a	none	·	·	conditional	stated
[153]	auditor	samples	exponent	n/a	none	none	·	conditional	stated
[158]	none	none	unstated	n/a	none	none	·	existential	stated
[57]	attacker	queries	exponent	stated	none	time	explicit	existential	stated
[195]	attacker	time	exponent	n/a	none	·	implicit	existential	stated
[199]	auditor	samples	budget	n/a	none	none	explicit	conditional	stated
[168]	attacker	corruption	budget	stated	algo class	none	explicit	conditional	stated
[140]	attacker	corruption	budget	n/a	none	none	explicit	conditional	stated
[137]	attacker	corruption	budget	stated	none	·	explicit	conditional	stated
[146]	attacker	corruption	budget	stated	none	perturbation	explicit	universal	stated
[72]	learner	time	poly	n/a	corruption	·	explicit	existential	stated
[70]	learner	samples	class	stated	none	algo class	explicit	conditional	stated
[190]	learner	samples	class	stated	none	algo class	explicit	universal	stated
[22]	learner	queries	budget	stated	corruption	time	implicit	universal	stated
[58]	auditor	time	poly	stated	corruption	none	explicit	conditional	stated
[157]	none	none	unstated	n/a	none	·	·	existential	none
[152]	attacker	queries	budget	n/a	none	time	implicit	empirical	stated
[186]	none	none	unstated	n/a	none	none	·	existential	none
[130]	auditor	time	poly	stated	perturbation	none	explicit	existential	stated
[109]	auditor	time	poly	absent	corruption	samples	explicit	conditional	stated
[105]	auditor	time	poly	stated	none	none	explicit	existential	stated
[12]	none	none	unstated	n/a	none	none	·	empirical	none
[68]	auditor	time	poly	n/a	none	none	explicit	empirical	stated
[160]	attacker	corruption	measured	stated	none	·	explicit	empirical	stated
[129]	none	none	unstated	n/a	none	none	·	empirical	stated
[154]	none	none	unstated	n/a	none	none	·	empirical	stated
[133]	mechanism	memory	exponent	stated	none	·	explicit	existential	stated
[9]	none	none	unstated	n/a	none	none	·	conditional	stated
[45]	attacker	time	exponent	n/a	time	none	explicit	conditional	stated
[76]	learner	time	exponent	n/a	none	none	·	conditional	stated
[17]	learner	algo class	class	n/a	none	·	explicit	conditional	stated
[35]	auditor	algo class	class	n/a	none	none	explicit	universal	stated
[38]	attacker	circuit size	budget	n/a	circuit size	algo class	explicit	existential	stated
[110]	attacker	memory	exponent	n/a	memory	queries	explicit	universal	stated
[21]	attacker	algo class	class	n/a	none	algo class	explicit	conditional	stated
[44]	learner	time	conditional	n/a	none	algo class	explicit	conditional	stated
[24]	attacker	memory	exponent	n/a	memory	time	explicit	conditional	stated
[60]	attacker	algo class	class	stated	none	·	explicit	existential	stated
[118]	attacker	time	poly	n/a	none	none	·	conditional	stated
[148]	attacker	time	poly	n/a	none	none	·	conditional	stated
[147]	attacker	time	poly	stated	none	none	explicit	existential	stated

Table 6, continued

#	Party	Unit	Tightness	Baseline	2nd party	2nd notion	Assessed	Quantifier	Antecedent
[62]	attacker	time	poly	stated	none	·	explicit	conditional	stated
[1]	attacker	circuit size	quasi-poly	stated	none	none	·	conditional	stated
[4]	mechanism	privacy budget	budget	n/a	none	none	·	empirical	stated
[37]	learner	time	poly	n/a	none	·	explicit	conditional	stated
[69]	learner	time	conditional	stated	none	algo class	explicit	conditional	stated
[14]	none	none	unstated	n/a	none	none	·	empirical	none
[87]	mechanism	memory	conditional	stated	time	none	explicit	conditional	stated
[161]	mechanism	time	budget	stated	none	·	explicit	universal	stated
[94]	learner	algo class	class	stated	none	queries	explicit	universal	stated
[89]	attacker	algo class	class	stated	algo class	none	explicit	conditional	stated
[36]	learner	time	conditional	n/a	none	·	explicit	conditional	stated
[15]	learner	time	conditional	n/a	none	none	explicit	conditional	stated
[5]	attacker	queries	exponent	stated	queries	none	explicit	existential	stated
[26]	attacker	time	exponent	stated	time	none	explicit	conditional	stated
[23]	attacker	circuit size	exponent	n/a	none	none	explicit	existential	stated
[11]	attacker	time	conditional	n/a	none	none	explicit	conditional	stated
[167]	none	none	unstated	n/a	none	none	·	empirical	stated
[103]	learner	circuit size	conditional	n/a	none	none	explicit	universal	stated
[104]	learner	algo class	class	n/a	none	·	explicit	universal	stated
[82]	learner	samples	exponent	stated	none	algo class	explicit	conditional	stated
[75]	learner	samples	exponent	absent	none	algo class	explicit	conditional	stated
[141]	learner	time	poly	stated	none	·	explicit	conditional	stated
[178]	learner	time	poly	stated	perturbation	·	explicit	conditional	stated
[120]	none	none	unstated	n/a	none	·	·	universal	stated
[123]	learner	time	poly	n/a	none	algo class	explicit	existential	stated
[194]	mechanism	algo class	class	stated	none	time	implicit	conditional	stated
[8]	learner	algo class	class	n/a	none	samples	explicit	universal	stated
[77]	auditor	algo class	class	n/a	privacy budget	samples	explicit	existential	stated
[39]	none	none	unstated	n/a	none	none	·	conditional	stated
[91]	learner	algo class	class	n/a	none	none	explicit	universal	stated
[169]	none	none	unstated	n/a	none	none	·	universal	stated
[3]	learner	algo class	class	n/a	none	time	explicit	conditional	stated
[67]	learner	algo class	class	n/a	none	queries	explicit	universal	stated
[119]	learner	time	poly	n/a	none	none	explicit	conditional	stated
[106]	attacker	time	poly	n/a	privacy budget	none	explicit	conditional	stated
[56]	learner	algo class	class	n/a	none	none	explicit	universal	stated
[175]	learner	algo class	class	n/a	none	none	explicit	universal	stated
[191]	learner	algo class	class	n/a	none	·	explicit	conditional	stated
[64]	none	none	unstated	n/a	none	none	·	conditional	stated
[2]	learner	algo class	class	n/a	none	samples	explicit	existential	stated
[42]	attacker	queries	exponent	n/a	queries	none	explicit	universal	stated
[189]	attacker	algo class	class	stated	algo class	none	explicit	conditional	stated
[73]	learner	samples	exponent	stated	none	algo class	explicit	conditional	stated
[102]	learner	parameters	budget	stated	none	time	explicit	existential	stated
[33]	attacker	time	exponent	n/a	time	none	explicit	conditional	stated
[138]	attacker	time	exponent	n/a	none	none	explicit	conditional	stated
[48]	attacker	circuit size	quasi-poly	stated	none	none	·	conditional	stated
[99]	attacker	circuit size	quasi-poly	stated	none	none	·	conditional	stated
[135]	learner	time	exponent	n/a	none	success prob	explicit	conditional	stated
[34]	learner	time	quasi-poly	n/a	none	algo class	explicit	existential	stated
[162]	learner	memory	budget	n/a	none	none	explicit	universal	stated
[20]	learner	time	conditional	stated	corruption	algo class	explicit	conditional	stated
[88]	learner	samples	exponent	stated	none	time	explicit	universal	stated
[116]	auditor	algo class	class	n/a	none	·	explicit	conditional	stated
[27]	mechanism	time	poly	n/a	time	none	explicit	conditional	stated
[107]	attacker	circuit depth	exponent	n/a	none	queries	explicit	existential	stated
[81]	attacker	time	poly	n/a	privacy budget	none	explicit	universal	stated
[83]	attacker	memory	exponent	stated	memory	·	explicit	universal	stated
[46]	learner	parameters	budget	n/a	none	none	explicit	conditional	stated

Table 6, continued

#	Party	Unit	Tightness	Baseline	2nd party	2nd notion	Assessed	Quantifier	Antecedent
[131]	learner	algo class	class	n/a	none	none	explicit	universal	stated
[198]	learner	time	poly	stated	none	samples	explicit	existential	stated
[41]	none	none	unstated	n/a	none	none	·	empirical	stated
[163]	mechanism	privacy budget	budget	stated	none	none	implicit	universal	stated
[85]	none	none	unstated	n/a	none	none	·	empirical	stated
[164]	attacker	samples	measured	stated	none	none	implicit	empirical	stated

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